Exploration: Solvolysis of Benzyl Tosylates

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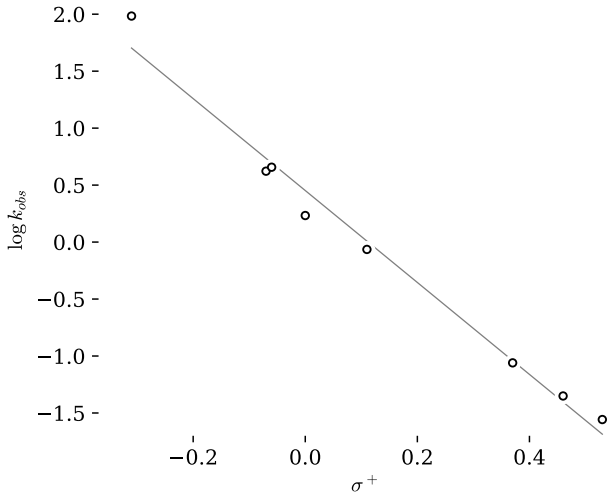
This exploration provides discussion for one of the cases presented in class meeting #14. In the class handout, you were asked to make a Hammett plot and come to a conclusion about the reaction mechanism. This document explores the data and alternative ways to interpret it by plotting and fitting it to mathematical models.[1]

Class Discussion: Curved Hammett Plots

This data comes from an investigation into the mechanism of solvolysis of benzyl tosylates in acetic acid.[2] If you read the paper, you will see that the authors interpret the data as representing two competing processes. One is a classic S_N2 solvolysis mechanism involving a moderate positive charge on the reaction centre at the transition state. The other is either a classic S_N1 process with a cationic intermediate or a very loose S_N2 transition state that could be described as a solvent-assisted S_N1 mechanism.[3]

The authors used σ^+ substituent parameters. Despite the S_N2 mechanism, the authors believe that there is a highly electron-deficient transition state that can accept electrons via resonance. Their hypothesis seems to be supported by the fact that a much better correlation is achieved by using these parameters rather than σ parameters. The data is collected in table 1

Figure 1 shows the plot of $\log k_{obs}$ vs. σ^+. The code used to create this plot and others is available as a *Python* notebook via Google Colab.



The plot had an excellent value of the Pearson correlation, R^2, but observe the plot of the residuals in figure 5. The linear model may not describe the

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[1] Read chapter 8.3 and 8.4.

[2] "Acetolysis reactivities of substituted benzyl and polycyclic arylmethyl p-toluenesulfonates. Correlations with SCF-π and CNDO MO methods." A. Streitwieser Jr., et. al., *J. Am. Chem. Soc.*, **1970**, *92*, 5141-5150 https://doi.org/10.1021/ja00720a025

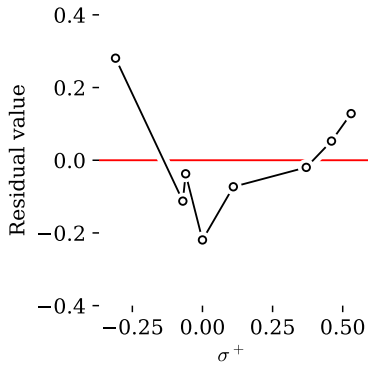
[3] See chapter 11.5 of the textbook

Table 1: Rates of solvolysis of benzyl toyslates in acetic acid at 40 °C. Data is from reference 2. ↓

Substituent	k /10^{-5} s^{-1}
p-CH_3	96.3
m-CH_3	4.54
p-F	4.19
H,	1.71
p-Cl	0.86
m-Cl	0.087
m-CF_3	0.0446
p-CF_3	0.0277

Figure 1: plot of $\log k_{obs}$ vs. σ^+. Do you see a bend in the plot? Streitwieser et. al. did. If we fit to a straight line we see that $\rho = -4.03 \pm 0.21$, $R^2 = 0.983$. ←

Figure 2: A plot of the residuals vs. σ^+ for the line fit in figure 1. We observe a systematic variation from the model, not random noise.↓



system.

The authors used the farther regions of the plot to fit a line for each system. They hypothesized that the electron-donating substituents favoured the transition state with more S_N1 character and that the electron-withdrawing substituents required a transition state consistent with the S_N2 mechanism. In figure 3 the same data is plotted but we used the four substituents with the most electron donating σ^+ values to fit one line and the three points at the other extreme to fit the other.

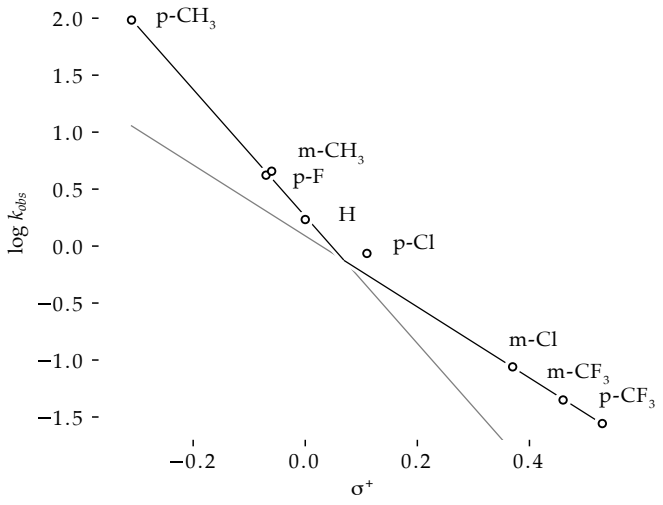


Figure 3: Plot of $\log k_{obs}$ vs. σ^+. The line fit of the upper four points gives $\rho = -5.57 \pm 0.22$, $R^2 = 0.997$ and the fit for the lower three points gives $\rho = -3.12 \pm 0.078$, $R^2 = 0.997$. ⟵

The ρ values of -3.1 and -5.6 indicate a substitution reaction with a large amount of positive character at the central carbon atom. In one case, we observe an S_N2 mechanism with positive character developing at the atom adjacent to the benzene ring. Recall that the hydrolysis of benzyl chloride in 50% water/acetone resulted in $\rho = -1.8$.[4] Neutral acetic acid is a weaker nucleophile than water, requiring a greater degree of departure of the leaving group in the transition state. Tosylate anion is a better leaving group than chloride ion, and this would require less involvement of the nucleophile in the transition state. We could thus propose that the transition state occurs with longer bonds to both the leaving group and nucleophile than in the case of hydrolysis of benzyl chloride.[5]

In the second case, we have a negative ρ value of greater magnitude. Substituents that stabilize the carbocationic nature of the transition state move the t.s. diagonally toward the extreme of S_N1. Both the bonds to the leaving group and the nucleophile are longer in the t.s. structure. Do we get all the way to something that could be described as solvent-assisted S_N1 or do we just get close? Either way, the alternate transition state has an even greater charge on the carbon atom.

We have two different transition states that are controlling the rate to product. One is the faster, given how much the substituents stabilize the

[4] See data tables 4 & 5 in the class handout for class meeting #12 and the corresponding Hammett plots in the accompanying discussion document.

[5] Start with the More O'Ferrall-Jencks plot for hydrolysis of benzyl chloride that we discussed in class meeting #12 and move the transition state as described here.

cationic nature of the reaction centre. Both transition states are cationic, and both are lowered in energy by electron-donating groups. But, the more extreme S_N1-like t.s. is affected to a greater degree. As we increase the electron-donating nature of substituents, this t.s. goes from representing the slower process to being the fastest route to the product. Both routes are occurring, but one is more predominant depending on the substituent. Sometimes it is not a choice between S_N2 and S_N1, and it is better described as a spectrum between these two extremes.[3,6]

[6] "The Correlation of Solvolysis Rates and the Classification of Solvolysis Reactions into Mechanistic Categories" S. Winstein, E. Grunwald, H.W. Jones, *J. Am. Chem. Soc.*, **1951**, *73*, 2700-2707. https://doi.org/10.1021/ja01150a078

Criticizing Someone's Hard Work

There is nothing wrong with this work presented in the paper by Streitwieser et. al.[2] ; however, I could not reproduce the ρ values reported from their version of the Hammett plot. The two slopes calculated by the authors were $\rho = -5.71$ and -2.33. Using the same data points and their reported values, my calculations gave $\rho = -5.57$ and -3.12.

The discrepancy could easily be explained by the fact that I am using data tables for the σ values that had been compiled in the 1990s[7] while these authors were using data from the 1960s. Small adjustments in the accepted data tables for substituent parameters have occurred over the years. The authors do not list the σ^+ values they used or document the source of the values or their methods for linear regression. That is why we document all of our calculations using interactive *Python* notebooks.

[7] "A survey of Hammett substituent constants and resonance and field parameters." C. Hansch, A. Leo, R. W. Taft, *Chem. Rev.*, **1991**, 91, 165-195. https://doi.org/10.1021/cr00002a004 and "Free Energy Relationships in Organic and Bio-organic Chemistry" by Andrew Williams, *The Royal Society of Chemistry, Cambridge*, **2003**, pp 258-277. https://doi.org/10.1039/9781847550927. We have access to the E-textbook version via the UPEI library.

The difference in the ρ values does not alter the conclusions, but it is always frustrating when I can't get the same numbers from the same data.

A Curved Plot

Two straight lines don't describe a curve. The authors recognize this but argue that they are using the two regions of the plot that best describe each route to determine the Hammett equation in each case. If you inspect the plot, you will see that the two lines are within an order of magnitude of each other for much of the plot. When we add the two rates together, we will see significant effects of both pathways in almost all data points across the plot.

For a start, let us add the two rates as determined from the two Hammett equations from figure 3. We cannot just add the $\log k_{obs}$ values calculated from the two sets of Hammett parameters. Recall that adding $\log k_{obs}$ values is the same as multiplying the two k_{obs} values together. We will calculate the k_{obs} value for each mechanism across the whole range of the plot, add those values together, and then take the $\log^{10}$ of the result. The math is documented in the *Python* notebook and the plot is presented in figure 4.

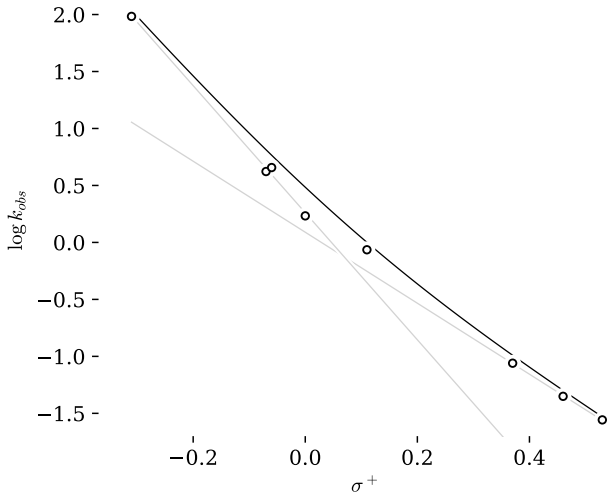
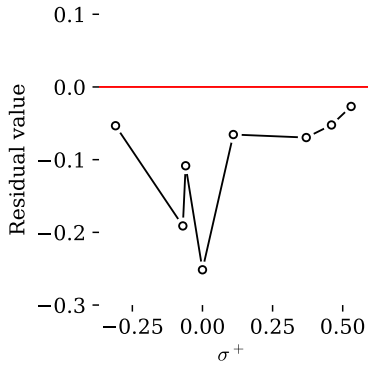


Figure 4: A plot of the total of the two different rate contributions as determined from the two line fits in figure 3 (each shown in grey). Observe that the curve tracks above the data. The calculated correlation between the curve and the data is $R^2 = 0.993$.
⟵

Figure 5: A he residuals vs. σ^+ for the line fit in figure 4. We observe a systematic variation from the model, not random noise. All the points are below the line, not randomly distributed.↓



The correlation coefficeint, R^2, is very good; however, if we plot the residuals we see a systematic variation from the model.

In this system, the two Hammett plots are close enough that we now see that both rates are significant in the total rate for almost the entire plot range. We should perform a curve fit to a mathematical model that considers this.

We will create a model that adds two Hammett plots together, one for each route to the product. Then, I can use the powerful statistical tools provided in *Python* to curve fit the data to the model. The result will be the Hammett parameters for the two reactions that add up to the total rate.

$$\log k_{total} = \log \left[10^{\left(\rho_1 \sigma^+ + \log c_1\right)} + 10^{\left(\rho_2 \sigma^+ + \log c_2\right)} \right]$$

The curve fit is presented in figure 6, along with the two Hammett plots based on the parameters determined for each process. For the more S_N1-like mechanism, we observed that $\rho_1 = -6.92 \pm 1.92$ and for the more S_N2-like mechanism we observed that $\rho_2 = -3.32 \pm 0.18$.[8] You will observe that the curve fit creates proposed Hammett parameters that result in the more S_N2-like mechanism being the predominant rate contributor in the plot. The standard deviation for the ρ_1 value of the S_N1-like mechanism is due to its lesser contribution and having only a single highly electron-donating system as a contributor that defines its parameters.

[8] Is the error reported a one standard deviation (66% confidence) or two (95% confidence)? Because the calculations are documented in the *Python* notebook, we can see that it is only one. You should double the error ranges for 95% confidence.

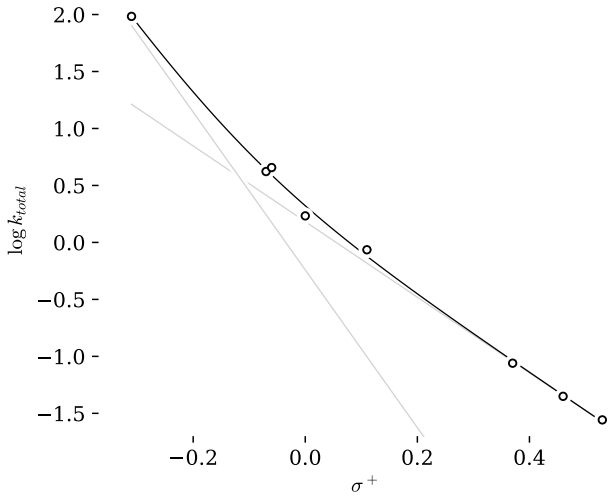
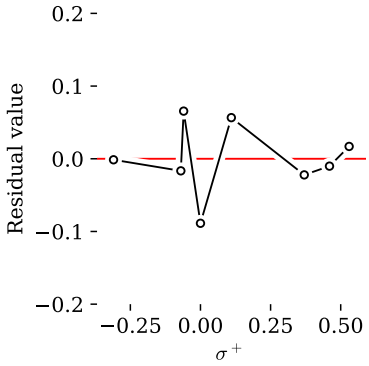


Figure 6: A plot of the total rate model vs. σ^+. The two Hammett plots that provide the best curve fit when added together are as follows...

$\log k_1 = (-6.92 \pm 1.92)\sigma^+ + (-0.24 \pm 0.22)$
$\log k_2 = (-3.32 \pm 0.18)\sigma^+ + (0.158 \pm 0.034)$
←

Figure 7: A plot of the residuals vs. σ^+ for the line fit in figure 6. We observe random noise, indicating some systematic variations from the model.↓



Regardless, the conclusion of the authors remains the same. We have two competing mechanisms, which are both S_N2; however, each has a different transition state structure. The more S_N2-like mechanism has a loose t.s. with a significant positive character on the central carbon, while the more S_N1-like mechanism has a very loose t.s. that might be better described as solvent-assisted S_N1. The extreme ρ_1 value indicates that the S_N1-like t.s. is poorly stabilized by the acetic acid solvent (compared to water).

Conclusion

Our analysis agrees with the authors' general conclusion. However, they presented their curved plot as two distinct regions that defined each mechanism. We have seen that it is better described as a plot that represents the more S_N2-like mechanism with a deviation due to increasing contribution from the more S_N1-like mechanism as we use more powerful electron-donating substituents. The calculated ρ value for the more S_N1-like mechanism is uncertain due to a lack of data, but we do observe that it has a high magnitude.[9]

We also came to see the benefits of documenting calculations and statistical analyses. Using *Python* notebooks to perform calculations and make the plots will enable others to understand exactly what was done rather than trying to guess what we did. The code used to create this plot and others is available via the *Python* notebook available via Google Colab.[10]

Looking Ahead

The ρ values are steep for S_N2. This observation can be explained by the greater importance of charge in a less polar solvent. However, we obtained the highly imprecise value of $\rho = (-6.92 \pm 1.92)$ for the electron-withdrawing

[9] The ρ value of -6.92 ± 1.92 is determined mostly by the two most electron-donating cases.

[10] All plots in this exploration were generated using the interactive *Python* notebook available via Google Colab at https://colab.research.google.com/github/blinkletter/4410PythonNotebooks/blob/main/Class_16/16_Class_Handout_Problems.ipynb.

region of the plot. That's a bit too steep for me. Are observing a complete change to an S_N1 mechanism between σ^+ values of 0 and -0.2?[11] We will be exploring the line between S_N1 and S_N2 again soon. Stay tuned.

[11] The authors quote a report by Hammond wherein a Hammett plot for benzyl tosylates (in aqueous conditions) show a continuous curve with the explanation being a steady change in transition state structure as electron density in the phenyl ring changes with substituents. That could be true, but figures 4 and 6 show plots that appear to be continuous curves but are the result of adding two straight lines together in a log-log plot. Data analysis is very important and can affect the conclusions.

Exploration Tips

This is the kind of work that I am hoping students do when writing their own exploration reports. It is short and more content would be expected, but there is a lot more to the paper than was presented here.

In this exploration, we took the data reported by Steitwieser et al.2 and replotted it as they did in their work. We matched their results. But we obtained an unsatisfactory fit when we added the two Hammett plots together. Using a more sophisticated model, we fit a model that included both Hammett plots and found the best fit was very uncertain about the steeper plot. Your eyes can fool you.[12]

[12] Your computer can fool you too. Sometimes even worse than your eyes.

The paper reports more than the rate constants for benzyl tosylates. The authors also investigated the reactivity of a series of polycyclic arylmethlyl tosylates. They used naphthyl groups and others in place of the phenyl groups. We cannot plot and analyze the Hammett plots for that series because there are few examples of reported substituent constants for these groups. The authors used both sets of rates to establish new σ^+ substituent constants for the polycyclic series. They then attempted to further support these new numbers by showing correlations between reported σ^+ values for benzyl groups and the new values for polycyclic groups with results from computational chemistry. They used low-level methods (it was the '70s after all).

So a student could easily extend this report by exploring how the authors established the new σ^+ values and used calculations to validate them.

Exploration Questions

Below is a set of questions I asked myself as I read the paper. I only explore the first of these questions in this exploration. If I had answered all of them, I believe this document would have been twenty pages long.

1. Hammett Plots: There is data for benzyl tosylate solvolysis in acetic acid at several temperatures. Rates were obtained by different methods.

   (a) I could replot the data, analyze it with modern methods and see if my conclusions agree with the authors.[13]

   (b) The authors report the methods used to perform the kinetic experiments: sealed tube, intermittent titration, stoppered flask, and conductivity. Each technique is briefly described and justified. I could look into each in more detail and explain how I believe the experiment was performed and the rate constants obtained.

[13] That is pretty much the only thing I explored in this document.

2. EYRING PLOTS: There is a table of thermodynamic data for the benzyl tosylate series. I could attempt to calculate these values from the temperature data mentioned above and see if I get the same numbers. I could plot Eyring plots using the reported numbers (or my calculated values) and see if there is an isokinetic point.

3. NEW SUBSTITUENT CONSTANTS: There is a table of rates for the polycyclic arylmethyl tosylates. The reactions were performed in acetic acid/chloroform mixtures due to low solubility.

   (a) The authors use an equation to extrapolate back to zero chloroform concentration and then show a linear plot to justify this extrapolation. I could plot the rest of the data and see if I agree (there is only one species in the data set that would provide a good second plot. Does it confirm the trend shown in the first plot?)

   (b) The authors then use their data and literature data to slide the rates for the polycyclic series into the rates for the benzyl series. By comparing how much the relative rates were changed by a substituted ring system, a new set of σ^+ values was proposed for substituted polycyclic arylmethyl groups. You could try to replicate their results and record the math of your method in detail.[14]

4. MO CALCULATIONS: At this point, we are only halfway through the paper. The second half discussed the molecular orbital energy calculations. That's out of my wheelhouse, so I'm not going there. However, a student interested in computational chemistry might explore and interpret their work. Replicate their results using the reported methods and then attempt to reinterpret their data using more modern methods on more powerful computers.[15]

As you can see, there was a lot more that could have been done. I could not explore it all in the 10-page body of work I expect in a student exploration report. Pick one, two or three things and dig into them. If your paper is short and doesn't have much to go on, you can choose another paper or explore one of the references in your paper.

[14] The authors gave no details about the exact math they used, but they did describe their approach. You'll have to do all the math on your own. Perhaps you will use a different approach compared to the authors. I expect you to get answers that are close to what was published. And I hope that you will document your calculations in a *Python* notebook.

[15] I would wager that your watch has a more powerful computer in it than was used in this work.